

GENERAL NOTES

1. GENERAL NOTES ON SHEET 2

3		
2		
1		
NO.	TITLE	DRAWING NO.

REFERENCE PLANS

SPECIAL NOTATIONS

PRELIMINARY DOCUMENT ISSUED TO SUPPORT DRAFT RFP. DOCUMENT IS UNDER DEVELOPMENT, AND NOT APPROVED.

RESERVATIONS:

1. POWER DEMAND OF BOW THRUSTER IS RESERVED PENDING BOW THRUSTER SIZING AND PERFORMANCE ANALYSIS.

STINFO DESIGNATION

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UNITED STATES COAST GUARD WASHINGTON, D.C. 20593

OFFICE OF NAVAL ENGINEERING

69 FT

HSCL

HOMELAND SECURITY CUTTER LIGHT
ELECTRIC PLANT LOAD ANALYSIS (EPLA)

	SIZE	DATE	USCG DRAWING NO.	REV
	A	3/11/2026	69-HSCL-300-001	0

GENERAL NOTES	
1.	INDIVIDUAL CONNECTED LOAD: FULL LOAD INPUT POWER (MEASURED IN KILOWATTS) OF ONE UNIT OF EQUIPMENT.
2.	STANDARD SERVICE LIFE ALLOWANCE IS 10%. DESIGN MARGIN IS 10%.
3.	ANALYSIS UTILIZES NOTIONAL COMPONENTS WHERE THE EXACT LOAD WAS UNKNOWN
4.	MULTIPLE OPERATING CONDITIONS, LISTED ON SHEET 7, WERE ANALYZED AND SUMMARIZED ON SHEET 5.
5.	TOTAL CONNECTED LOAD: THE PRODUCT OF INDIVIDUAL CONNECTED LOAD AND QUANTITY OF UNITS
6.	THE ELECTRIC PLANT STATUS SUMMARY SHEETS DEMONSTRATE THE ELECTRIC PLANT CAPABILITIES INCLUDING SERVICE LIFE ALLOWANCE MARGIN (SLA).
7.	ALL CONDITIONS SHOWN ON EPLA ARE FOR MANNED CONDITIONS. THERE ARE NO UNMANNED CONDITIONS SHOWN ON THIS CALCULATION.
8.	THE FOLLOWING LOADS ARE CONSIDERED TO BE EMERGENCY LOADS: 1. EMERGENCY LIGHTS 2. NAVIGATION LIGHTS 3. FIRE DETECTION AND FIXED FIREFIGHTING SYSTEMS 4. COMMUNICATION EQUIPMENT 5. NAVIGATION EQUIPMENT 6. GENERAL ALARM SYSTEM

REVISIONS					
REV	ITEM	SWBS	DESCRIPTION	DATE	INITIAL
0	-	-	PRELIMINARY RELEASE FOR DRAFT RFP	3/11/2026	-

Electric Plant Status Summary:	
Total Shore Power Capacity (kW) (Single Cord)	83.14
Number of Ship Service Generators	2.00
Generator Rating, (kW) each	65.00
Total Installed Generator Capacity, (kW) each	130.00
Total Installed Capacity, (kW)	130.00

AC Load Summary - Summer						
SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (kW)	TRANSIT (kW)	ATON (kW)	
200	Propulsion Plant	9.88	6.88	0.58	0.58	
300	Electric Plant	74.12	14.89	10.43	21.89	
400	Command & Surveillance	3.79	0.76	1.32	2.37	
500	Auxiliary Systems	43.15	8.50	10.42	24.59	
512	Heating Ventilation & Air Cond (less heaters)	16.48	14.83	14.83	14.83	
512H	HVAC Heaters	30.21	0.00	0.00	0.00	
600	Outfitting	8.93	0.62	0.79	0.79	
583	Deck Equipment	0.00	0.00	0.00	0.00	
Total AC Load		186.57	46.47	38.37	65.05	
Total AC Load w/ 20% Margin +SLA		223.89	55.76	46.04	78.06	
Shore Power Online		-	1.00	0.00	0.00	
Number of Generators Online		-	0.00	1.00	1.00	
Number of Inverters Online		-	0.00	0.00	0.00	
Excess Electrical Plant Capacity		-	36.67	26.63	-0.05	
Generator Loading		-	56%	59%	100%	
Excess Electrical Plant Capacity w/ 20% Margin + SLA		-	27.38	18.96	-13.06	
Generator Loading w/ 20% Margin + SLA		-	67%	71%	120%	

AC Load Summary - Winter						
SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (19°F) (kW)	TRANSIT (-20°F) (kW)	ATON (0°F) (kW)	ICEBREAKING (-20°F) (kW)
200	Propulsion Plant	9.88	6.88	0.58	0.58	0.58
300	Electric Plant	74.12	14.89	10.43	21.89	10.43
400	Command & Surveillance	3.79	0.61	1.27	2.37	2.23
500	Auxiliary Systems	43.15	8.50	10.42	23.38	10.36
512	Heating Ventilation & Air Cond (less heaters)	16.48	1.27	1.27	1.27	1.27
512H	HVAC Heaters	30.21	10.17	27.19	21.48	27.19
600	Outfitting	8.93	1.88	2.05	2.05	2.05
583	Deck Equipment	0.00	0.00	0.00	0.00	0.00
Total AC Load		186.57	44.19	53.21	73.02	54.11
Total AC Load w/ 20% Margin + SLA		223.89	53.02	63.85	87.63	64.93
Shore Power Online		-	1.00	0.00	0.00	0.00
Number of Generators Online		-	0.00	1.00	2.00	1.00
Number of Inverters Online		-	0.00	0.00	0.00	0.00
Excess Electrical Plant Capacity		-	38.95	11.79	56.98	10.89
Generator Loading		-	53%	82%	56%	83%
Excess Electrical Plant Capacity w/ 20% Margin + SLA		-	30.12	1.15	42.37	0.07
Generator Loading w/ 20% Margin +SLA		-	64%	98%	67%	100%

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AC SUMMARY

SIZE	DATE	USCG DRAWING NO.	REV
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Electric Plant Status Summary:	
Total Shore Power Capacity (kW) (Single Cord)	
Number of UPS	2.00
UPS Voltage (VDC)	24.00
NAV Equipment Voltage (VDC)	12.00
UPS Rating, (kW) each	5.00
Total Installed UPS Capacity, (kW) each	10.00

DC Load Summary - Summer						
SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (kW)	TRANSIT (kW)	ATON (kW)	
200	Propulsion Plant	1.44	0.29	0.86	1.15	
300	Electric Plant	1.06	0.47	0.66	0.66	
400	Command & Surveillance	1.17	0.21	0.44	0.54	
500	Auxiliary Systems	9.36	1.98	0.26	3.80	
512	Heating Ventilation & Air Cond (less heaters)	0.46	0.28	0.28	0.28	
512H	HVAC Heaters	0.00	0.00	0.00	0.00	
600	Outfitting	0.65	0.00	0.00	0.00	
583	Deck Equipment	0.00	0.00	0.00	0.00	
Total DC Load		14.14	3.22	2.50	6.43	
Total DC Load w/ 20% Margin +SLA		16.97	3.87	3.00	7.72	
Shore Power Online		-	0.00	0.00	0.00	
Number of UPS Online		-	1.00	1.00	2.00	
Excess UPS Capacity		-	1.78	2.50	3.57	
UPS Loading		-	64%	50%	64%	
Excess UPS Capacity w/ 20% Margin + SLA		-	1.13	2.00	2.28	
UPS Loading w/ 20% Margin + SLA		-	77%	60%	77%	

DC Load Summary - Winter						
SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (19°F) (kW)	TRANSIT (-20°F) (kW)	ATON (0°F) (kW)	ICEBREAKING (-20°F) (kW)
200	Propulsion Plant	1.44	0.29	0.86	1.15	0.86
300	Electric Plant	1.06	0.47	0.66	0.66	0.66
400	Command & Surveillance	1.17	0.21	0.44	0.54	0.54
500	Auxiliary Systems	9.36	1.98	0.22	3.80	0.22
512	Heating Ventilation & Air Cond (less heaters)	0.46	0.24	0.24	0.24	0.24
512H	HVAC Heaters	0.00	0.00	0.00	0.00	0.00
600	Outfitting	0.65	0.00	0.00	0.00	0.00
583	Deck Equipment	0.00	0.00	0.00	0.00	0.00
Total DC Load		14.14	3.19	2.42	6.40	2.53
Total DC Load w/ 20% Margin + SLA		16.97	3.82	2.91	7.67	3.03
Shore Power Online		-	0.00	0.00	0.00	0.00
Number of UPS Online		-	1.00	1.00	2.00	1.00
Excess UPS Capacity		-	1.81	2.58	3.60	2.47
UPS Loading		-	64%	48%	64%	51%
Excess UPS Capacity w/ 20% Margin + SLA		-	1.18	2.09	2.33	1.97
UPS Loading w/ 20% Margin +SLA		-	76%	58%	77%	61%

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DC SUMMARY

SIZE	DATE	USCG DRAWING NO.	REV
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SWBS GROUP	EQUIPMENT NAME	QTY	EM / NE	PHASE	KW	VOLTAGE	AMPERES	FREQUENCY	INDIVIDUAL CONNECTED LOAD (KW)	TOTAL CONNECTED LOAD (KW)	LOAD CATEGORY (SWBS)	WINTER SHORE LOAD FACTORS	WINTER SHORE KW	SUMMER SHORE LOAD FACTORS	SUMMER SHORE KW	WINTER TRANSIT LOAD FACTORS	WINTER TRANSIT KW	SUMMER TRANSIT LOAD FACTORS	SUMMER TRANSIT KW	WINTER ICEBREAKING LOAD FACTORS	WINTER ICEBREAKING KW	WINTER (0°F) ATON LOAD FACTOR	WINTER (0°F) ATON KW	SUMMER ATON LOAD FACTORS	SUMMER ATON KW	WINTER EMERGENCY LOAD FACTORS	WINTER EMERGENCY KW	SUMMER EMERGENCY LOAD FACTORS	SUMMER EMERGENCY KW
500	HOT WATER TANK	1.00	NE	1.00	18.00	120.00	10.00	60	18.00	18.00	533.00	0.10	1.80	0.10	1.90	0.50	8.00	0.50	8.00	0.50	8.00	0.50	8.00	0.50	8.00	0.00	0.00	0.00	0.00
300	UNINTERRUPTIBLE POWER SUPPLY	2.00	NE	1.00	11.00	240.00	45.83	60	11.00	22.00	313.00	0.35	7.70	0.35	7.70	0.27	5.94	0.27	5.94	0.27	5.94	0.27	5.94	0.27	16.50	0.00	0.00	0.00	0.00
512H	REHEATER (BRIDGE)	2.00	NE	1.00	2.75	120.00	22.92	60	2.75	5.50	512H	0.34	1.85	0.00	0.00	0.90	4.95	0.00	0.00	0.90	4.95	0.65	3.58	0.00	0.00	0.00	0.00	0.00	0.00
512H	DUCT HEATER (REFRESHMENT)	2.00	NE	1.00	2.50	120.00	20.83	60	2.50	5.00	512H	0.34	1.68	0.00	0.00	0.90	4.50	0.00	0.00	0.90	4.50	0.60	3.00	0.00	0.00	0.00	0.00	0.00	0.00
512H	DUCT HEATER (AUX MACHINERY ROOM)	2.00	NE	1.00	2.50	120.00	20.83	60	2.50	5.00	512H	0.34	1.68	0.00	0.00	0.90	4.50	0.00	0.00	0.90	4.50	0.65	3.25	0.00	0.00	0.00	0.00	0.00	0.00
512H	UNIT HEATER (ENGINE ROOM)	2.00	NE	1.00	2.50	120.00	20.83	60	2.50	5.00	512H	0.34	1.68	0.00	0.00	0.90	4.50	0.00	0.00	0.90	4.50	0.80	4.00	0.00	0.00	0.00	0.00	0.00	0.00
512H	UNIT HEATER (AUX MACHINERY ROOM)	2.00	NE	1.00	2.50	120.00	20.83	60	2.50	5.00	512H	0.34	1.68	0.00	0.00	0.90	4.50	0.00	0.00	0.90	4.50	0.75	3.75	0.00	0.00	0.00	0.00	0.00	0.00
400	CCTV	1.00	NE	1.00	2.88	120.00	24.00	60	2.88	2.88	439.00	0.20	0.58	0.20	0.58	0.40	1.15	0.40	1.15	0.70	2.02	0.70	2.02	0.70	2.02	0.00	0.00	0.00	0.00
512H	REHEATER (MESS, BERTHING)	1.00	NE	1.00	2.00	120.00	16.67	60	2.00	2.00	512H	0.34	0.67	0.00	0.00	0.90	1.80	0.00	0.00	0.90	1.80	0.90	1.80	0.00	0.00	0.00	0.00	0.00	0.00
300	PILOTHOUSE RECEPTACLE	8.00	NE	1.00	1.80	120.00	15.00	60	1.80	14.40	320.00	0.10	1.44	0.10	1.44	0.10	1.44	0.10	1.44	0.10	1.44	0.10	1.44	0.10	1.44	0.00	0.00	0.00	0.00
600	BRIDGE HEATED WINDOWS	1.00	NE	1.00	6.30	120.00	52.50	60	6.30	6.30	625.00	0.20	1.26	0.00	0.00	0.20	1.26	0.00	0.00	0.20	1.26	0.20	1.26	0.00	0.00	0.00	0.00	0.00	0.00
500	STEERING SYSTEM HPU	1.00	NE	3.00	2.24	230.00	5.82	60	2.24	2.24	561.00	0.00	0.00	0.00	0.50	1.12	0.50	1.12	0.50	1.12	0.50	1.12	0.50	1.12	0.00	0.00	0.00	0.00	0.00
512H	CONVECTION HEATER (T/S HEAD)	2.00	NE	1.00	0.50	120.00	4.17	60	0.50	1.00	512H	0.34	0.34	0.00	0.00	0.90	0.90	0.00	0.00	0.90	0.90	0.90	0.90	0.00	0.00	0.00	0.00	0.00	0.00
512H	UNIT HEATER (LAZARETTE)	1.00	NE	1.00	1.00	120.00	8.33	60	1.00	1.00	512H	0.34	0.34	0.00	0.00	0.90	0.90	0.00	0.00	0.90	0.90	0.68	0.67	0.00	0.00	0.00	0.00	0.00	0.00
300	MESS RECEPTACLE (DOUBLE)	4.00	NE	1.00	1.80	120.00	15.00	60	1.80	7.20	320.00	0.10	0.72	0.10	0.72	0.10	0.72	0.10	0.72	0.10	0.72	0.10	0.72	0.10	0.72	0.00	0.00	0.00	0.00
512H	REHEATER (FWD BERTHING)	1.00	NE	1.00	0.75	120.00	6.25	60	0.75	0.75	512H	0.34	0.25	0.00	0.00	0.90	0.68	0.00	0.00	0.90	0.68	0.65	0.49	0.00	0.00	0.00	0.00	0.00	0.00
512H	FAN COIL UNIT HEATER (BRIDGE)	2.00	NE	1.00	0.36	230.00	1.55	60	0.36	0.71	512H	0.34	0.24	0.00	0.00	0.90	0.54	0.00	0.00	0.90	0.54	0.90	0.54	0.00	0.00	0.00	0.00	0.00	0.00
200	MPDE BATTERY CHARGER (24VDC)	2.00	NE	1.00	1.44	120.00	12.00	60	1.44	2.88	223.00	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.00	0.00	0.00	0.00
300	SSDG BATTERY CHARGER (24VDC)	2.00	NE	1.00	1.44	120.00	12.00	60	1.44	2.88	313.00	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.20	0.58	0.00	0.00	0.00	0.00
200	GEARBOX CONTROL POWER - PORT & STBD	2.00	NE	1.00	0.48	24.00	20.00	DC	0.48	0.96	252.00	0.20	0.19	0.20	0.19	0.60	0.58	0.60	0.58	0.60	0.58	0.80	0.77	0.80	0.77	0.00	0.00	0.00	0.00
300	COFFEE/MAKER	1.00	NE	1.00	1.72	120.00	14.33	60	1.72	1.72	651.00	0.20	0.34	0.20	0.34	0.30	0.52	0.30	0.52	0.30	0.52	0.30	0.52	0.30	0.52	0.00	0.00	0.00	0.00
300	MESS RECEPTACLE (SINGLE)	2.00	NE	1.00	2.40	120.00	20.00	60	2.40	4.80	320.00	0.10	0.48	0.10	0.48	0.10	0.48	0.10	0.48	0.10	0.48	0.10	0.48	0.10	0.48	0.00	0.00	0.00	0.00
512H	CONVECTION HEATER (ANCHOR)	1.00	NE	1.00	0.50	120.00	4.17	60	0.50	0.50	512H	0.34	0.17	0.00	0.00	0.90	0.45	0.00	0.00	0.90	0.45	0.45	0.23	0.00	0.00	0.00	0.00	0.00	0.00
512	VENTILATION FAN, SUPPLY (ENGINE ROOM)	2.00	NE	3.00	2.24	230.00	5.62	60	2.24	4.47	512.00	0.10	0.45	0.90	4.03	0.10	0.45	0.90	4.03	0.10	0.45	0.10	0.45	0.90	4.03	0.00	0.00	0.00	0.00
500	AUX SW COOLING PUMP	1.00	NE	1.00	0.56	120.00	4.66	60	0.56	0.56	524.00	0.50	0.28	0.50	0.28	0.70	0.39	0.70	0.39	0.70	0.39	0.70	0.39	0.70	0.39	0.00	0.00	0.00	0.00
500	FRESHWATER COOLING PUMP	1.00	NE	1.00	0.56	120.00	4.66	60	0.56	0.56	532.00	0.50	0.28	0.50	0.28	0.70	0.39	0.70	0.39	0.70	0.39	0.70	0.39	0.70	0.39	0.00	0.00	0.00	0.00
500	BALLAST WASHDOWN / FIRE PUMP	2.00	NE	1.00	3.73	230.00	9.36	60	3.73	7.45	320.00	0.10	0.75	0.20	0.15	0.37	0.10	0.15	0.20	0.15	0.37	0.10	0.15	0.20	0.15	0.00	0.00	0.00	0.00
300	FWD BERTHING RECEPTACLE	2.00	NE	1.00	1.80	120.00	15.00	60	1.80	3.60	320.00	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.00	0.00	0.00	0.00
300	ENGINE ROOM RECEPTACLE (DOUBLE)	2.00	NE	1.00	1.80	120.00	15.00	60	1.80	3.60	320.00	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.10	0.36	0.00	0.00	0.00	0.00
512	FAN COIL UNIT (BRIDGE)	2.00	NE	1.00	0.20	230.00	0.86	60	0.20	0.39	512.00	0.90	0.35	0.90	0.35	0.90	0.35	0.90	0.35	0.90	0.35	0.90	0.35	0.90	0.35	0.00	0.00	0.00	0.00
300	BATTLE LANTERN	7.00	EM	1.00	0.08	24.00	3.50	DC	0.08	0.56	332.00	0.40	0.24	0.40	0.24	0.60	0.35	0.60	0.35	0.60	0.35	0.60	0.35	0.60	0.35	1.00	0.00	0.00	0.00
512H	FAN COIL UNIT HEATER (MESS AND SR5)	1.00	NE	1.00	0.36	230.00	1.55	60	0.36	0.99	512H	0.34	0.12	0.00	0.00	0.90	0.32	0.00	0.00	0.90	0.32	0.34	0.34	0.00	0.00	0.00	0.00	0.00	0.00
200	MAIN ENGINE CONTROL - PORT & STBD E-BOX	2.00	NE	1.00	0.24	24.00	10.00	DC	0.24	0.48	252.00	0.20	0.10	0.20	0.10	0.60	0.29	0.60	0.29	0.60	0.29	0.80	0.38	0.80	0.38	0.00	0.00	0.00	0.00
600	REFRIGERATOR	1.00	NE	1.00	0.90	120.00	7.50	60	0.90	0.90	651.00	0.30	0.27	0.30	0.27	0.30	0.27	0.30	0.27	0.30	0.27	0.30	0.27	0.30	0.27	0.00	0.00	0.00	0.00
300	WEATHER DECK RECEPTACLE	1.00	NE	1.00	1.80	120.00	15.00	60	1.80	1.80	320.00	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.60	1.08	0.60	1.08	0.00	0.00	0.00	0.00
300	DECK HEAD RECEPTACLE	1.00	NE	1.00	1.80	120.00	15.00	60	1.80	1.80	320.00	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.00	0.00	0.00	0.00
300	STATEROOM (2 PERSON) RECEPTACLE	1.00	NE	1.00	1.80	120.00	15.00	60	1.80	1.80	320.00	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.10	0.18	0.00	0.00	0.00	0.00
512H	FAN COIL UNIT HEATER (FWD BERTHING)	1.00	NE	1.00	0.20	230.00	0.86	60	0.20	0.20	512H	0.34	0.07	0.00	0.00	0.90	0.18	0.00	0.00	0.90	0.18	0.90	0.18	0.00	0.00	0.00	0.00	0.00	0.00
512H	FAN COIL UNIT HEATER (WORKSHOP)	1.00	NE	1.00	0.20	230.00	0.86	60	0.20	0.20	512H	0.34	0.07	0.00	0.00	0.90	0.18	0.00	0.00	0.90	0.18	0.90	0.18	0.00	0.00	0.00	0.00	0.00	0.00
512	FAN COIL UNIT (MESS AND SR5)	1.00	NE	1.00	0.20	230.00	0.86	60	0.20	0.20	512.00	0.90	0.18	0.90	0.18	0.90	0.18	0.90	0.18	0.90	0.18	0.90	0.18	0.90	0.18	0.00	0.00	0.00	0.00
400	HANDHELD RADIO SYSTEM	2.00	EM	1.00	0.10	12.00	8.33	DC	0.10	0.20	441.00	0.10	0.02	0.10	0.02	0.40	0.08	0.40	0.08	0.40	0.08	0.14	0.70	0.14	1.00	0.20	1.00	0.20	0.00
512	VENTILATION FAN, SUPPLY (REFRESHMENT)	1.00	NE	1.00	0.13	24.00	5.50	DC	0.13	0.13	512.00	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.00	0.00	0.00	0.00
512	VENTILATION FAN, EXHAUST (T/S HEAD)	1.00	NE	1.00	0.13	24.00	5.50	DC	0.13	0.13	512.00	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.90	0.12	0.00	0.00	0.00	0.00
512	FAN COIL UNIT (SR FWD)	1.00	NE	1.00	0.11	230.00	0.49	60	0.11	0.																			